

Qualification

Thapar Institute of Engineering and Technology

B.Tech in Computer Engineering — CGPA: 8.61/10

2024 – 2028 (Expected)

Patiala, India

Modern Senior Secondary School

Class XII – PCM with Computers — Percentage: 92.6%

2022 – 2024

Patiala, India

Technical Skills

Languages: C++, Python, JavaScript, HTML, CSS, MATLAB, R

Frameworks/Libraries: React.js, Node.js, FastAPI, MediaPipe, dnd-kit, TensorFlow, OpenCV, Pandas, NumPy, scikit-learn

Cloud / Databases: MongoDB, Firebase, SQL, Netlify, Render, Supabase, Vercel

Developer Tools: Git, Docker, Linux, Postman

Coursework: Data Structure and Algorithm, Operating Systems, Relational Database Concepts, Object Oriented Programming, Artificial Intelligence, Computer Networks

Experience

Auraflo (TIET Ventures Lab)

Mar 2025 – Sep 2025

AI/ML Intern

Remote

- Engineered a real-time pose detection pipeline processing live streams at **30-45 FPS** using TensorFlow Lite and OpenCV.
- Developed state-transition logic with confidence-based filtering for accurate movement quality evaluation, achieving **95%** accuracy across **500+** test sessions.

Projects

AtomQuest — Compliance Portal | *Next.js 14, React, Supabase, Vanilla CSS* (GitHub | Live)

May 2026

- Designed and implemented a serverless-safe caching layer in Next.js 14, reducing export generation overhead by **60%** on Vercel deployments.
- Built a workflow-driven compliance escalation engine and an end-to-end encrypted communication module with client-side encryption and decryption.

Spatial Console — Gesture-Controlled Structural Simulation | *React, Three.js, MediaPipe* (GitHub)

Feb. 2026

- Integrated gesture-driven interactions at **45–60 FPS** into a real-time browser-based 3D system using React, Three.js, and MediaPipe.
- Implemented a graph-based structural integrity engine using **BFS traversal** to detect instability and simulate cascading failures.

Code Vault — Competitive Programming Platform | *React, Node.js, MongoDB, Tailwind CSS* (GitHub | Live)

Jan. 2026

- Deployed a full-stack live coding competition platform handling **30+ concurrent teams** with **2-second** auto-refresh, **3-second** leaderboard polling, and warnings for tab switching to preserve contest integrity.
- Developed contest management tooling for challenge distribution, monitoring, evaluation, and automated result generation.

Positions of Responsibility

LEAD Society (TIET)

Oct 2024 – Present

Technical Secretary

Patiala, India

- Built the official LEAD Society website: lead2025.netlify.app using HTML, CSS, and JavaScript; maintained the site and deployment.
- Mentored **100+** members in web development and Git workflows; spearheaded technical events for a **500+** student community.

Modern Senior Secondary School

2022 – 2023

Vice Art & Cultural Captain

Patiala, India

- Organized 4 **school-wide** events and anchored 3 ceremonial programs, reaching a cumulative audience of **2,000+** students and faculty.

Achievements

- Solved **500+** DSA problems on LeetCode, GFG, and CodeChef, with focus on medium-hard problems across trees, graphs, DP, and sliding window.
- Merit III** Scholarship awarded for scoring 95.4 percentile in JEE Mains.
- District Gold Medalist & School Chess Champion:** Recognized for excellence in competitive strategy and performance.